

Organization and Management Exam Notes

ME 708 · IV-I (BEL, BEX, BCT, BEI) · TU, IOE · tiers over 29 papers, 2065–2082 BS

5. Management Information System

5 hrs · asked in **25 of 29 papers** · 8 syllabus topics · **248 marks, 10.5% of the archive** → about 6 of every 80 marks: one 8-mark question, split 3+5 or 4+4.

READ THIS FIRST

MIS questions almost always open with **“define MIS / why is MIS needed?”** (17 papers) and add one of three second halves:

1. **Types of information systems** (TPS, MIS, DSS, ESS) and DSS vs MIS (7 papers) → one table in §5.3.
2. **Information support for functional areas** (5 papers) or **at different levels** → one pyramid figure in §5.3 answers both.
3. **Planning and decision making** (7 papers), **computers and MIS**, **database** and **networking** systems → §5.2 and §5.4.

The 2065–2068 papers asked “hierarchy of information needs” and “information architecture”; only S.K. Joshi’s textbook covers them, and they are summarized from it.

Tier chips count **papers out of 29**: **TOP n/29** 8+ **HOT n/29** 4–7 **PIN n/29** 1–3 inside an 8-mark question **LOW n/29** 1–3, short note only syllabus, no PYQ yet never set. Bold year = Regular exam.

5.1 Data, Information and MIS

Define MIS. Why is MIS needed? Its functions and importance **2 3 4 5**

TOP 17/29 82 Ba, 81 Ba, 80 Bh, 79 Ba, 76 Ch, 76 Ash, 74 Ash, 73 Shr, 71 Ch, 70 Ch, 70 Asa, 69 Ch, 68 Ch, 68 Ba, 67 Asa, 66 Bh, 65 Shr · “needs for MIS” short note in 68 Ch, 66 Bh, 65 Shr; also the second half of 73 Ch and 71 Ch (filed with case studies)

- The three words:
 - **Management**: directing an enterprise toward objectives (planning, organizing, coordinating, controlling).
 - **Information**: processed, orderly selected data that gives knowledge for decisions.
 - **System**: interacting procedures forming an organized whole.
- **MIS**: “a system of obtaining, abstracting, storing and analysing data to produce information for planning, controlling and decision making, given to managers at the time they can best use it.”
- Henry C. Lucas: “a set of organized procedures which, when executed, provide information to support decision making.”
- Uses computing and communication technology; draws on internal and external sources; covers past, present and projected activity.

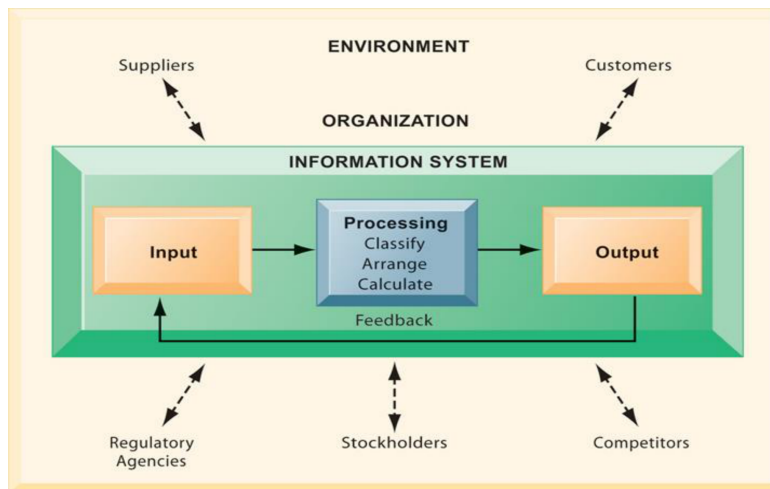
Functions

1. **Data capture**: from internal and external sources.
2. **Processing**: classify, calculate, compare, summarize.
3. **Storage and retrieval**.
4. **Determining information needs**: what, how much, when, for whom.
5. **Evaluation and abstraction**: judge reliability, edit, condense.
6. **Dissemination**: right information, right manager, right time.
7. **Prediction, planning, control** support.

Adhikari’s four, with his questions

- **Determination** of needs: what? how much? how, when, by whom?
- **Evaluation**: how much confidence can be placed in it?
- **Abstraction**: editing and reducing incoming information.
- **Dissemination**: right information, right manager, right time.

The information system inside the organization, inside its environment



The organization's information system within its environment.

Need for MIS

- Every manager decides on information; more complex firms → more data than a person can handle.
- **Production managers:** production, labour, machine, overhead costs; capacity.
- **Marketing managers:** sales trends, market research, selling costs, new products.
- **Personnel managers:** turnover, absenteeism, skills, wage levels, labour market.
- **Why MIS in addition to task software (79 Ba):** payroll, billing, inventory packages each serve one task; MIS **integrates** their data into summary and exception reports across the firm, for decisions no single package supports.

Importance

- The "heart" of the firm: timely, relevant information for **decisions**.
- **Measures performance against goals** → control.
- Supports **planning**, forecasting, coordination across departments.
- Reduces uncertainty and cost of information; keeps it current.
- Stores confidential information securely for future use.
- Competitive advantage: faster response to market.
- Eg a bank's MIS flags branches whose loan defaults exceed target; an airline's reservation data sets fares by demand.
- Also:
 - Brings new facts to management's notice.
 - Supplies information at a **reasonable cost**.
 - Focuses on decisions with an attractive **benefit/cost ratio**.
 - Supplies reliable, logical data for every management function.

Data vs information vs knowledge vs wisdom; how data and information are used in an office 4

PIN 3/29 82 Ba, 78 Bh, 75 Ash

- **Data:** raw, unprocessed facts and figures; no meaning alone. Eg 45, 62, 38.
- **Information:** data processed into a meaningful, useful form. Eg “average daily sales this week: Rs 48,000, down 12%”.
- **Knowledge:** information combined with experience and understanding: knows *how*. Eg “sales drop every exam month”.
- **Wisdom:** applying knowledge with judgement to decide well: knows *why* and *what is best*. Eg “run a student discount in exam months”.
- DIKW pyramid (right): each level adds context and value.

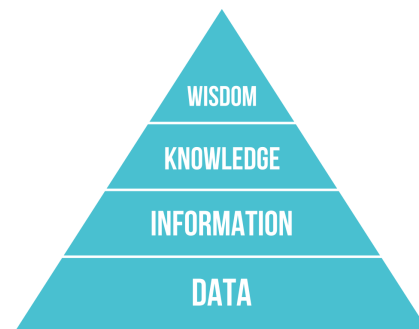
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Adhikari's contrast

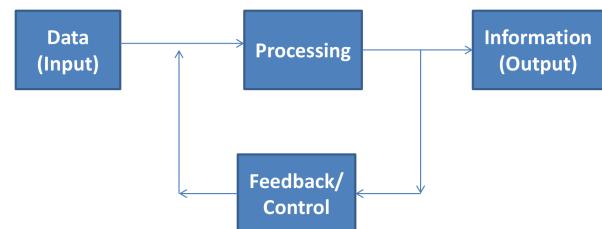
- **Data:**
 - Uninterpreted raw statement of facts.
 - Non-random symbols (words, values, figures) representing what has happened.
 - From research or observation. Eg payroll, bank statement.
- **Information:**
 - Facts organized to give the desired knowledge.
 - Data interpreted and understood by the recipient.
 - Can mean different things to different people.

Qualities of good information

- Accurate; timely; complete; concise.
- Relevant; at the right frequency; understandable.
- Worth its cost: costs less than its value.



DIKW pyramid. Wikimedia Commons (Longlivetheux), CC BY-SA 4.0.



Data becomes information through processing, with feedback.

Use in an office (78 Bh, 75 Ash)

- **Data** recorded daily: attendance, invoices, receipts, stock movements, letters.
- Processed into **information**: payroll, monthly accounts, stock reports, pending-file lists.
- Used for: paying staff, ordering supplies, answering customers, monitoring work, reporting to management, planning budgets.

5.2 MIS for Planning, Decision Making and Management Levels

Role of information in the planning process and in decision making; information system is vital for planning and control 4 5 6 8

HOT 7/29 76 Ch, 76 Ash, 68 Ch, 68 Ba, 67 Asa, 66 Bh, 65 Shr · short note "information system for planning process" in 68 Ba

Information system for planning

- Planning answers: **where are we? where do we want to go? how do we get there? when? who? how much will it cost?**
- MIS supplies:
 - Past performance: where we are.
 - Forecasts of demand, costs, market, technology: where to go.
 - Resource data: how, who, at what cost.
- Long-range (3–5 year) plan → yearly operating plan → individual targets: MIS tracks each.
- Planning is continuous; MIS keeps the plan a "living document" by feeding actual results back.

Vital for control (68 Ch)

- Control = compare actual with plan → correct. Without timely information there is nothing to compare.
- Exception reports highlight only deviations (management by exception).

Information system for decision making

- Decision = choosing the best alternative; more and better information → better choice.
- Steps supported:
 1. Identify the problem: exception reports.
 2. Generate alternatives: data, models.
 3. Evaluate: what-if analysis, DSS.
 4. Choose the best.
 5. Monitor: feedback reports.
- **Strategic:** future-oriented, long range, unstructured: new plant, new market.
- **Tactical:** short term: budgets, fund flow, personnel problems, product improvement.
- **Operational:** structured, routine: inventory, scheduling, allocating workers.

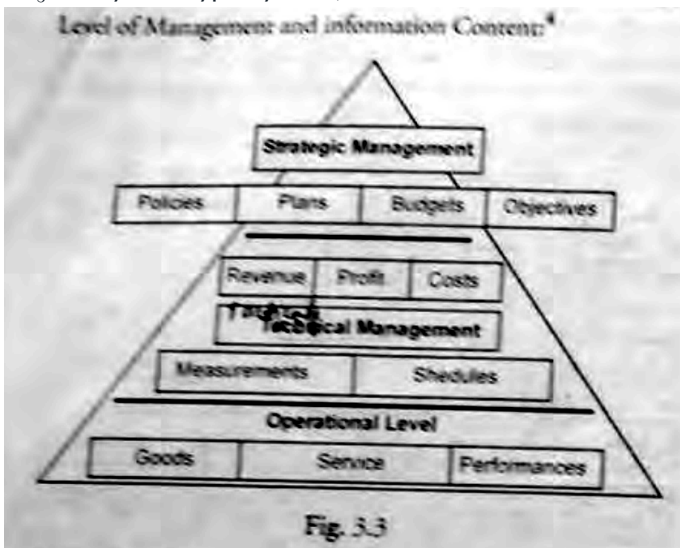
Value of MIS in the planning process (76 Ch, 76 Ash)

- **Accurate forecasts** of demand, cost, prices.
- **Realistic targets** based on past performance.
- **Scenario analysis:** what if sales fall 10%?
- **Early warning** when plans slip.
- **Coordination** of departmental plans from one shared database.

Hierarchy of information needs; organizational structure and MIS; MIS at different levels of management 5 8

HOT 4/29 80 Ba, 68 Ch, 66 Bh, 65 Shr · from S.K. Joshi, Ch. 3.1.1

- Managers at different levels need **different kinds** of information at **different intervals**: the **hierarchy of information needs**. MIS is a pyramid matching the organization's levels.
- Operational (lower)**: day-to-day production of goods and services.
 - Detailed, internal, precise, historical data.
 - Hourly or daily: raw material needs, work schedules, output vs schedule.
- Tactical (middle)**: puts top's plans into operation.
 - Summaries: schedules, revenue, profit, cost, other economic measures.
 - Weekly or monthly; used to set controls and allocate resources.
- Strategic (top)**: corporate policy, growth, survival.
 - Aggregated, future-oriented, much of it **external** (competitors, suppliers, consultants, media).
 - Quarterly or yearly: are goals met, is the firm profitable, what should change.
- Bottom of the pyramid: **transaction processing** data feeding all levels.
- For "MIS at different levels" (80 Ba), draw the pyramid in §5.3: system type by level, across the functional areas.



Level of management and information content: policies, plans, budgets at the top; revenue, profit, costs, measurements, schedules in the middle; goods, services, performance at the bottom. S.K. Joshi, Fig. 3.3.

Why the hierarchy is necessary (68 Ch)

- Each level gets **only what it needs**: no overload at top, no gaps below.
- Detail is condensed upward (reports), direction flows downward (plans).
- Matches decision type to information type.

Organizational structure and MIS

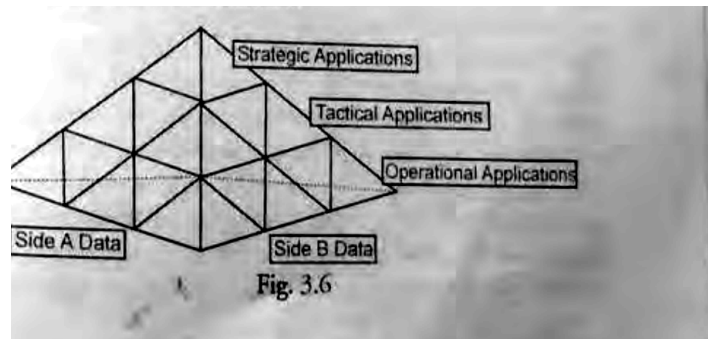
- Nature of the firm (profit or not, goods or services, single site or global) decides its information needs and systems.
- MIS draws on organization concepts: structure, communication, power, decision making, motivation, leadership.
- Adhikari: MIS is a pyramid of **four levels** of information resources: strategic, tactical, operational control, transaction processing. Since decisions are specific to levels, so are information needs.

Level	Scope	Source	Frequency
Strategic	Wide, summary, future	Mostly external	Quarterly, yearly
Tactical	Summaries, trends	Internal	Weekly, monthly
Operational	Narrow, detailed, current	Internal	Hourly, daily

Information architecture 3 4

PIN 2/29 68 Ba, 65 Shr · from S.K. Joshi, Ch. 3.2

- From Greek *archos* (chief) + *tekton* (carpenter): design and structure.
- Information architecture = **how the parts of an information system interconnect and communicate**: the general **flow** of information, how **databases interrelate**, and how applications make access easier for managers.
- Also (AJ Sir): organizing, structuring and labelling content so users find information and complete tasks.
- **Anthony's triangle**: strategic, tactical, operational applications stacked by level.
- Add **divisions** (production, marketing, accounting) → a two-dimensional structure (figure below).
 - Each division keeps its unique data.
 - Common events (a sale) create **shared data** → a **shared corporate database**.
- Traditional bureaucracy: information flows up, over, down; divisions rarely talk directly. Good architecture removes these barriers.



Anthony's triangle extended across divisions: side A and side B data meet in shared data. S.K. Joshi, Fig. 3.6.

5.3 Types of Information Systems and Functional Support

Classification of information systems (TPS, MIS, DSS, ESS) and their support to managers; DSS vs MIS; EIS for top managers 2 3 4 5 6

HOT 7/29 82 Bh, 80 Bh, 79 Ba, 78 Bh, 75 Ash, 71 Ch, 69 Ch

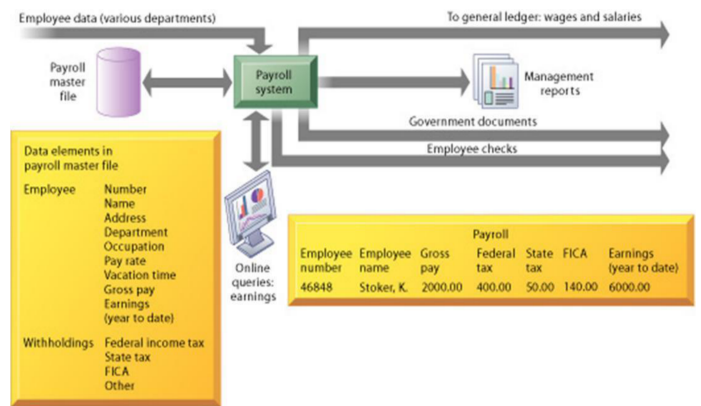
- **Information system**: hardware, software, data, people and procedures that produce information for day-to-day, short-range and long-range activities.

TPS

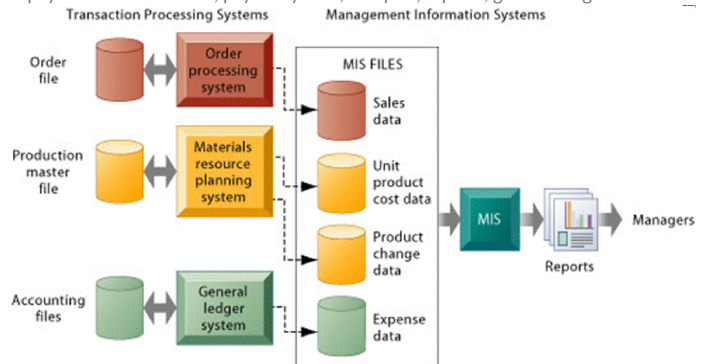
- Captures and processes data from **day-to-day transactions**: deposit, payment, order, reservation. Serves the operational level.
- Two modes:
 - **Online**: each transaction processed at once (deposits, reservations); most TPS today.
 - **Batch**: transactions grouped and run later (paychecks, invoices); efficient for bulk.

MIS: three kinds of report

- Accurate, timely, organized information to decide, solve problems, supervise, track progress; reports regularly, so also **management reporting system**.
- **Detailed**: confirms each transaction. Eg detailed order report.
- **Summary**: totals, tables, graphs. Eg inventory summary.
- **Exception**: only what is out of range. Eg items below reorder level.



A payroll TPS: master file, payroll system, cheques, reports, general ledger.



TPS files (orders, materials, ledger) feed MIS files and reports to managers.

Basis	TPS	MIS	DSS	ESS / EIS
Level served	Operational	Middle (management)	Middle, professionals	Top (strategic)
Inputs	Transactions, events	Summarized transaction data	Transaction data + models, external data	Aggregate internal + external data
Processing	Sort, list, update, batch or online	Simple models, routine reports	Interactive, what-if analysis	Interactive, graphics, drill-down
Outputs	Detailed reports, lists	Summary and exception reports	Decision analysis	Projections, dashboards
Decisions	Structured	Structured, recurring	Semi-structured	Unstructured
Example	Payroll, order entry, billing	Annual budgeting, sales by region	Contract cost analysis, production scheduling	5-year operating plan

DSS vs MIS (78 Bh, 75 Ash)

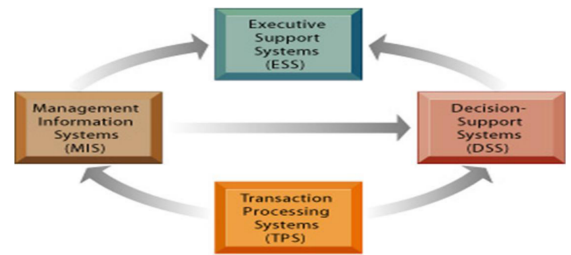
Basis	MIS	DSS
Purpose	Routine reporting, control	Support a specific decision
Problems	Structured	Semi-/unstructured
Output	Fixed, periodic reports	Ad hoc, interactive analysis
User role	Receives reports	Manipulates data and models
Data	Internal, past	Internal + external, future
Flexibility	Low	High (what-if)

How production managers use DSS in addition to MIS (82 Bh)

- MIS tells **what happened**: output, scrap, machine downtime reports.
- DSS asks **what if**: best production schedule and product mix (linear programming), make-or-buy, capacity expansion, inventory reorder levels, maintenance timing, with simulations of cost and delay.

Significance of EIS for top managers (79 Ba)

- One-screen dashboards of key indicators; drill-down to detail.
- External data: competitors, markets, economy.
- Fast, easy to use without technical skill.
- Supports strategic decisions and early warning; less dependence on staff reports.



TPS feeds MIS and DSS; all feed ESS.

Support to managers

- TPS: records every transaction; the data source for all others.
- MIS: regular detail, summary and exception reports → control (also called management reporting system).
- DSS: models + data to decide non-routine problems.
- ESS: strategic view for top managers.
- **Expert system**: knowledge base + inference rules imitate an expert (Eg medical or fault diagnosis).
- Modern software integrates them (ERP): firms consolidate TPS, MIS and DSS into one integrated system.

DSS: data from inside and outside

- **Internal**: sales, manufacturing, inventory, finance.
- **External**: interest rates, population trends, construction costs, raw material prices.
- Eg a voyage-estimating DSS runs models over ship, port, fuel and charter-cost files.

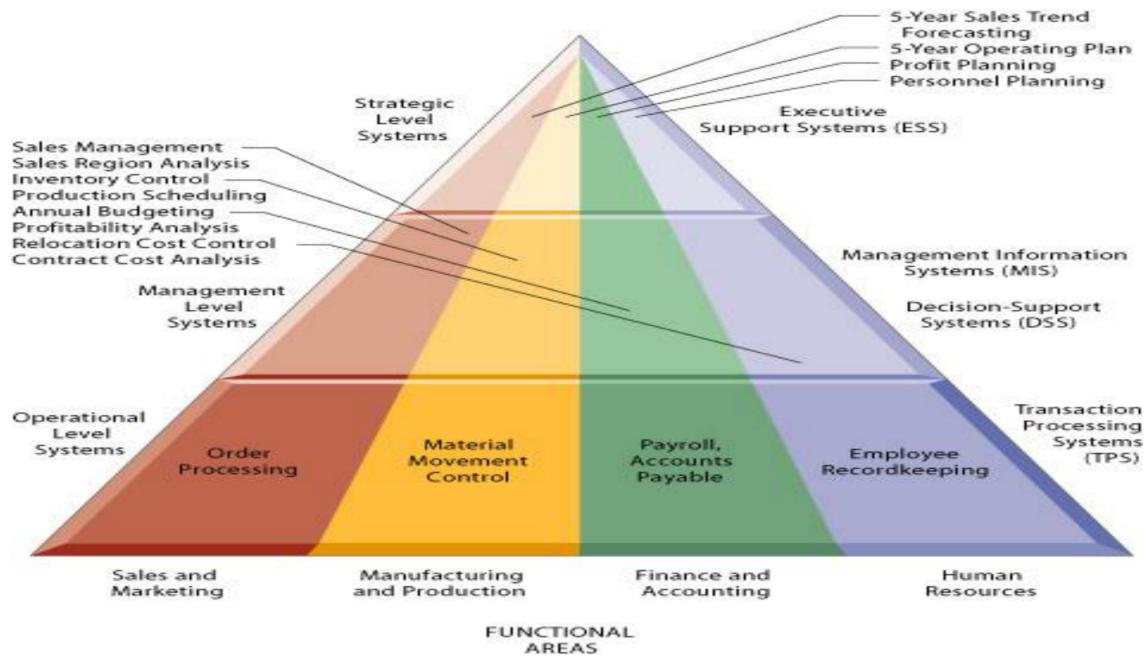
Expert system: how it reasons

1. User describes the situation.
2. Inference rules make logical judgements.
3. Knowledge base holds experts' knowledge and experience.
4. System gives advice or a decision.

Information support for functional areas of management; how MIS is used at different levels 4 5 6

8

HOT 5/29 74 Ch, 73 Shr, 72 Ka, 70 Ch, 70 Asa · 80 Ba's "MIS at different levels" is answered by the same figure



Four functional areas (columns) crossed with three levels (rows): each cell is the information system that supports it.

Sales and marketing

- TPS: order processing, invoicing.
- MIS: sales by region, salesperson, product; sales management.
- DSS: pricing, sales region analysis, promotion planning.
- ESS: 5-year sales trend forecast, market share.

Manufacturing and production

- TPS: material movement, machine control, purchase orders.
- MIS: inventory control, production scheduling reports, quality and scrap.
- DSS: production planning, relocation cost, capacity.
- ESS: 5-year operating plan, new plant decisions.

Finance and accounting

- TPS: payroll, accounts payable and receivable, general ledger.
- MIS: annual budgeting, cost reports.
- DSS: profitability, contract cost analysis, investment appraisal.
- ESS: profit planning, long-term financing.

Human resources

- TPS: employee record keeping, attendance.
- MIS: turnover, absenteeism, training reports, appraisal history.
- DSS: compensation analysis, manpower allocation.
- ESS: personnel (manpower) planning.

5.4 Computers, Databases, Networks and Organizing Information Systems

Computers and MIS; role of computers in MIS; what is a website 2 4

HOT 4/29 81 Bh, 76 Ash, 74 Ash, 68 Ba

- Computers are the **means** of modern MIS, not MIS itself: a computer alone does not make work efficient; identify the management problem first.
- **Before computers** information was: too late, incomplete, costlier than its worth, unfocused, irrelevant, rigid.
- **Role of computers:**
 - Fast, accurate processing of huge data.
 - Storage and quick retrieval.
 - Reports in any format, on demand.
 - Communication across sites.
 - Models and simulation for decisions.
- **Quick-response systems:**
 - **Online processing:** manager interacts directly with the computer.
 - **Real-time processing:** system runs alongside the activity and guides it (Eg ATM, airline booking).
 - **Time sharing:** many users share one computer at high speed.
- Two uses: **data (transaction) processing** with fixed reports, and **end-user computing** (managers, staff using spreadsheets, queries) with flexibility.
- **Relationship:** MIS defines what information is needed; the computer produces it quickly and cheaply. Computer = tool, MIS = purpose.
- **Website** (74 Ash): a set of linked web pages under one domain name, hosted on a server and reached through the internet. **verified/added**
 - Publishes information to customers.
 - Collects orders and feedback (data).
 - Gives staff remote access to reports.
- **Computer-aided advertising** (69 Ch, short note **LOW 1/29** 69 Ch): **verified/added**
 - Computer graphics and design for ads.
 - Targeted online, email and social media ads.
 - Analytics of clicks and conversions.

Database information system 4 5

HOT 4/29 81 Ba, 68 Ba, 66 Bh, 65 Shr · short note in 68 Ba

- **Database:** organized collection of related data serving many applications at once, appearing to be in one place.
- Kinds:
 - **Personal database:** an individual's contacts, schedules, notes.
 - **Corporate database:** large, shared, integrity critical.
- **DBMS:** software to create, maintain and query it (Oracle, MySQL); applications extract data without their own files.
- Components:
 - Data dictionary: describes the data.
 - Architecture: relations between data.
 - The objects described; rules for manipulation.
- Advantages: no redundancy, consistency, sharing, security, easy reporting.
- Examples:
 - A bank's customer database used by accounts, loans, ATM and mobile banking.
 - A college's student database used by admission, exams, library, fees.

Networking information system 4 5

PIN 2/29 67 Asa, 65 Shr · short note in 65 Shr

- Telecommunication network linking computers and terminals so users at different locations share data and resources.
- Components: computers, terminals, channels, communication processors, software; common rules = **protocols**; analog or digital signals.
- Types:
 - **By area:** LAN (a building), MAN (a city), WAN (country or world).
 - **By users:** internet (public), intranet (staff only), extranet (staff plus partners, suppliers).
- Needs a flexible, standard network architecture that grows with the business.
- Advantages: real-time sharing, distributed offices, e-commerce, email, lower cost.
- Eg bank branches and ATMs on one network; NEA billing linked across offices.

How information systems can be organized properly 4 6

PIN 3/29 81 Bh, 76 Ash, 71 Shr · short note in 71 Shr

- **Complete centralization:** one central IS department holds hardware, software, data and MIS personnel; serves all user departments.
 - + control, standards, economy of scale, security.
 - – slow response, users feel distant, bottleneck.
- **Complete decentralization:** each department has its own systems and MIS staff.
 - + responsive, users own their systems.
 - – duplication, incompatible data, higher cost.
- **Hybrid (distributed):** central database and standards, local processing and staff. Most common today.
 - + control of shared data with local responsiveness.
 - – needs clear rules on who owns what.

Other means of organizing it properly

- **Steering committee** of senior managers sets IS priorities.
- **CIO** (chief information officer) at top level.
- Clear **policies** for data ownership, access and security.
- **Standards** for hardware, software, data formats.
- **User training** and a help desk.
- Regular **audit and review** of the system.

Roles of an organized MIS department

- Hardware administration; database administration.
- Software development and support.
- End-user support and training.
- Computing standards; corporate data access.
- Advocacy: promotes IS use across the firm.

Evolution of MIS

syllabus, no PYQ yet

1. Manual ledgers and punch cards.
 2. Electronic data processing (1950s–60s): record keeping.
 3. MIS reports (1960s–70s).
 4. DSS (1970s): models, operations research.
 5. EIS / ESS (1980s).
 6. PCs and end-user computing: graphical, user-friendly interfaces.
 7. Networks, ERP: integrated systems.
 8. Internet, cloud and business intelligence.
- Driven by: growth of management theory, management science, new production methods and structures, and computers.

5.5 PYQ Mapping

Theory questions, covered in this document

- **Define MIS; need, functions, importance; MIS beyond task software** → §5.1. 17 papers; plus the second half of 73 Ch and 71 Ch, which the Detailed PYQ files under case studies.
- **Data, information, knowledge, wisdom; use in an office** 82 Ba 4, 78 Bh 4, 75 Ash 4 → §5.1.
- **Information for planning, decision making, control** → §5.2. 7 papers.
- **Hierarchy of information needs; MIS at levels** 80 Ba 8, 68 Ch 5, 66 Bh 5, 65 Shr 4; **information architecture** 68 Ba 3, 65 Shr 4 → §5.2.
- **Types of IS, DSS vs MIS, DSS for production, EIS** → §5.3. 7 papers.
- **Information support for functional areas** → §5.3. 74 Ch 8, 73 Shr 4, 72 Ka 8, 70 Ch 6, 70 Asa 5
- **Computers and MIS; website; computer-aided advertising** → §5.4. 81 Bh, 76 Ash, 74 Ash, 69 Ch, 68 Ba
- **Database IS** 81 Ba 5, 68 Ba 5, 66 Bh 5, 65 Shr 4; **networking IS** 67 Asa 5, 65 Shr 4; **organizing IS** 81 Bh 6, 76 Ash 6, 71 Shr 4 → §5.4.
- **Evolution of MIS:** syllabus, never set → §5.4.

Numerical content

- This chapter has no numerical component.

Sources and corrections

- Hierarchy of information needs, information architecture, planning and the quick-response systems come from S.K. Joshi, Ch. 3 (pp 99–135), read through `tools/ocr_page.py`; the lecture notes do not cover them.
- Adhikari p283 expands DSS as “design support system”; it is **decision** support system. Adhikari p296 heads the ESS slide but describes an **expert** system; both are given separately above.

- The Detailed PYQ's "hierarchy of needs" in 65 Shr sits in the MIS question and is read as hierarchy of information needs, as 66 Bh words it.

Figures

- Only diagrams an answer would draw are kept; no photos, clip art or decorative charts.
- Cropped: Adhikari's notes (pp277, 281, 291, 293, 299, 300) and S.K. Joshi Figs. 3.3, 3.6 (book photographs, rendered grayscale with auto-contrast).
- DIKW pyramid: Wikimedia Commons (Longlivetheux, CC BY-SA 4.0).

Abbreviations, chips and month codes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
JSC	joint stock company	MNC	multinational company
HR, HRM	human resource (management)	OB	organizational behaviour
MBO	management by objectives	CEO, MD	chief executive officer, managing director
AGM	annual general meeting	ESOP	employee share ownership plan
MoA, AoA	memorandum / articles of association	OCR	Office of the Company Registrar
SEBON	Securities Board of Nepal	NEPSE	Nepal Stock Exchange
IPO	initial public offering	PAN, VAT	permanent account no., value added tax
NEA	Nepal Electricity Authority	NOC	Nepal Oil Corporation
QC	quality control	R&D	research and development
OR	operations research	ROI	return on investment
CIM	computer integrated manufacturing	FMS	flexible manufacturing system
CAD, CAM	computer aided design / manufacturing	CAPP	computer aided process planning
CNC	computer numerical control	AGV	automated guided vehicle
MIS	management information system	IT	information technology
SaaS	software as a service	BPO	business process outsourcing
CSR	corporate social responsibility	POSDCoRB	plan, organize, staff, direct, coordinate, report, budget

Short	Means	Short	Means
no.	number	info	information
vs, Vs	versus	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	min, max	minimum, maximum
Ch.	chapter	§	section

Chip	Means (29 papers, 2065–2082 BS)
TOP n/29	asked in 8 or more papers
HOT n/29	asked in 4 to 7 papers
PIN n/29	asked in 1 to 3 papers, inside an 8-mark question
LOW n/29	asked in 1 to 3 papers, as a 4-mark short note only
syllabus, no PYQ yet	in the syllabus, never examined
4+4	marks as printed; several chips = the question recurred at different marks
verified/added	researched and added beyond the lecture notes
80 Bh	bold year = Regular exam; plain = Back exam

Months: Ba = Baishakh, Jth = Jestha, Asa = Ashad, Shr = Shrawan, Bh = Bhadra, Ash = Ashwin, Ka = Kartik, Mng = Mangsir, Po = Poush, Ma = Magh, Ch = Chaitra. **Ash and Asa are different months.** **Sources:** AJ Sir = AJ Sir's lecture decks; Adhikari = Bikash Adhikari, *Organization and Management* slide notes; Joshi = S.K. Joshi, *Organization and Management*, A.K. Books.